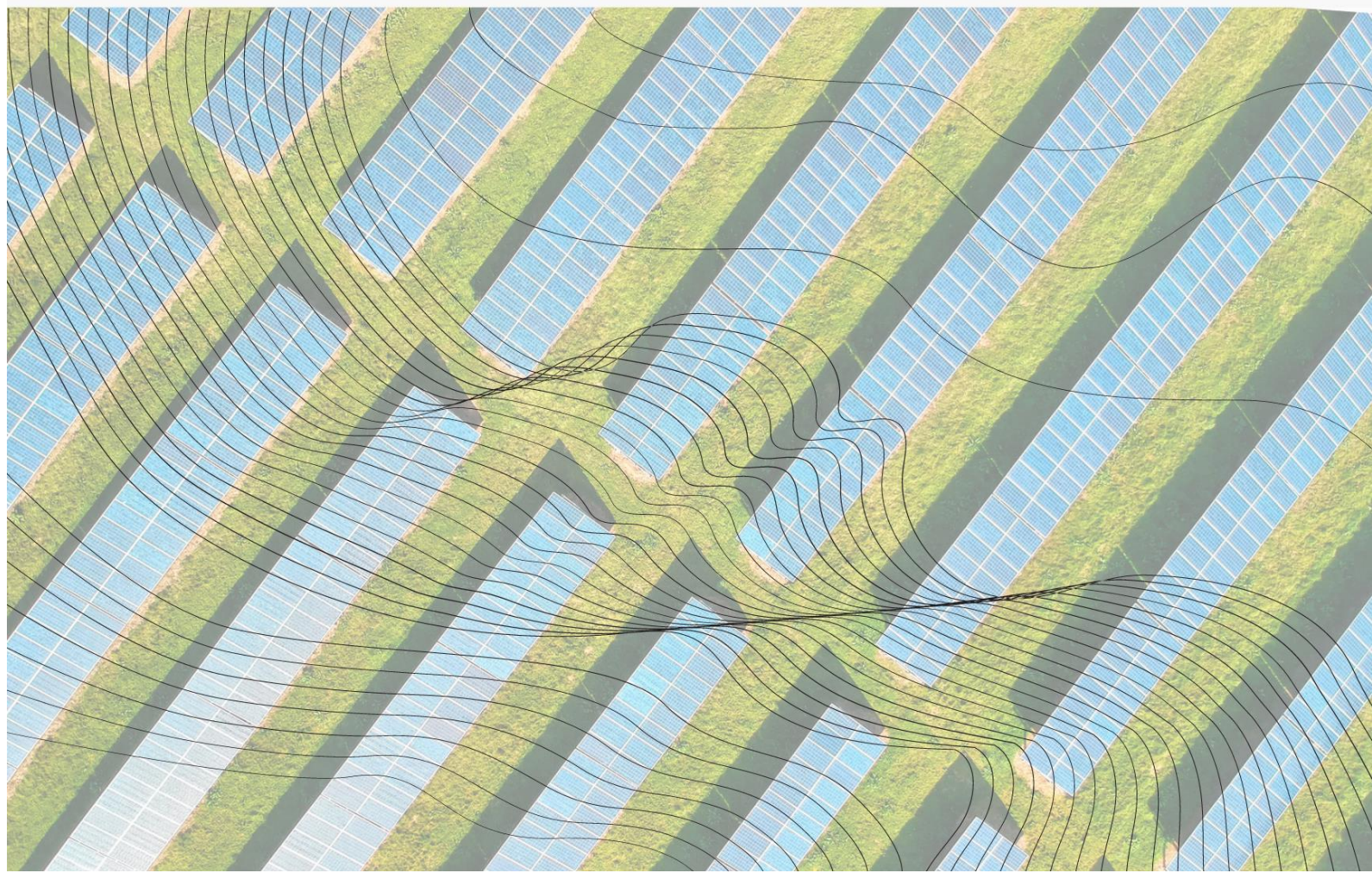




HANNON ARMSTRONG



Introduction and Company Overview

Hannon Armstrong is a Real Estate Investment Trust (REIT) that is solely dedicated to investing in climate change solutions. Their investment thesis states, “We will earn better risk-adjusted returns by investing on the right side of the climate change line” (Hannon Armstrong, 2020). By providing capital to companies that work in the fields of energy efficiency, renewable energy, and sustainable market infrastructure, Hannon Armstrong seeks to yield high returns on its investments. Based in Annapolis, Maryland, the company invests in a diversity of projects across the United States, including onshore wind farms, large scale solar facilities, and sustainable infrastructure. Their goal for these projects is to generate consistent, predictable cash flows while simultaneously offsetting carbon emissions. Hannon Armstrong manages \$6.1 billion in assets and has a market cap of \$3.9 billion. The firm is publicly traded under the ticker HASI and engages heavily with both the federal and local governments. Their unique business model and environmental stance has brought the company success in recent years; HASI currently trades at five times its 2013 IPO price and the company reported net incomes of \$81 million in 2019.

Company Leadership



Jeffrey Eckel- Chairman & CEO since 2000. His strong background in renewable energy infrastructure is one of the main reasons for the company's strong pivot to the climate positive space.



Jefferey Lipson- Executive Vice President and CFO. Prior to joining Hannon Armstrong, Mr. Lipson's entire career was in the banking industry.



J. Brendan Herron- Executive Vice President. Mr. Herron has over 25 years of experience in structuring technology investments and focuses on company strategy and growth initiatives.

Company Timeline

The firm that is contemporarily known as Hannon Armstrong was founded in 1981 by Mike Hannon and Greg Eden under the name Eden Hannon Goodwin & Company. The founders recognized an opportunity to utilize a tax-exempt lease by financing capital equipment upgrades for state and local governments. Their first investment was in telecommunications equipment for a Virginia police department. In 1983, Jon Armstrong joined the company. He brought his experience in federal contracting and would eventually earn eponymic status within the firm. 1987 was a pivotal year for the company as it made its first transaction in renewable energy: the refinancing of a solar system in California. Throughout the 80's and 90's, Hannon Armstrong would continue to operate in both the renewable energy and telecommunications sectors. In 1996, it completed its first Energy Service Performance Contract, which enables federal agencies

to procure energy savings and facility improvements with no up-front capital costs or special appropriations from Congress. This type of contract would become a staple of the firm's business model.

In 2000, Jeff Eckel returned as CEO of the company after spending 11 years working in Europe to introduce renewable, high-efficiency power systems in emerging markets. Throughout the early 2000s, Hannon Armstrong would transition its focus more heavily towards sustainable energy systems. Though it would continue to finance telecommunication infrastructure (including a fiberoptic cable into the Arctic Circle), the increased prevalence and profitability of green energy investment would eventually drive Hannon Armstrong to narrow its focus and emphasize the mitigation of climate change. In 2007, management was bought out and Hannon Armstrong became 100% employee owned. However, the period of employee ownership was short-lived, as Hannon Armstrong debuted as a public company on the New York Stock Exchange in 2013. Since then, Hannon Armstrong has rebranded itself as a firm that invests exclusively in climate change solutions and has built out the three-pronged business model that it employs today (Hannon Armstrong Timeline, 2020).

Industry Analysis

REIT Industry Overview

Hannon Armstrong operates as a Real Estate Investment Trust (REIT), though its emphasis on renewable energy infrastructure is unique amongst REITs. REITs were established through the Real Estate Investment Trust Act of 1960. They function as a vehicle through which investors can fund investment in a property (or more likely, a collection of properties) and yield a return when the property generates income. Put simply, REITs own and operate income-generating property, the profits of which are used to pay out investors. Because many REITs are publicly traded on major exchanges, investors can buy in for as little as the price of a single share. Many people describe REITs as real estate mutual funds, which is conceptually true except for one big difference: REITs are traded like stocks. A single REIT is a single company that likely manages many operations. Because of this, ownership in a share of a REIT could be compared to ownership of a bond: the share (bond) holder is entitled to profits yielded by the operations of the company proportional to the number of shares that they own. Mutual funds, on the other hand, are bought and sold through investment firms and are composed of many different companies.

All REITs fall into one of three categories:

- *Equity REITs* own a set of residential or commercial properties and generate returns by collecting interest from tenants. Properties are typically purchased with mortgages or corporate bonds.
- *Mortgage REITs* do not own and operate real estate. Rather, they lend to real estate owners directly or acquire existing loans; revenue comes from interest received on these loans.
- *Specialty REITs* are relatively new. This category encompasses REITs that do not fit the above two descriptions. These REITs operate in niche categories and are often highly specialized. Examples of specialty REIT operations include correctional facilities, data centers, or billboards (Krewson-Kelly & Thomas 2016).

A REIT's designated category does not exclude it from partaking in income-generating activities that fall under another category. In fact, many REITs differentiate amongst their competitors by hybridizing the above three categories in their operations. The operational category that derives the most income for the firm becomes the firm's designated category. Hannon Armstrong's income-generating activities include a variety of investments that yield returns in different ways, including interest and rent collection. However, because all operations take place in the niche space of investing in projects that offset global climate change, Hannon Armstrong is qualified as a specialty REIT.

By law, REITs must pay out at least 90% of taxable income in the form of shareholder dividends. Doing so gives the REIT a tax-advantaged status. Most REITs pay annual dividends of 2-3%, though some pay over 10%. Other qualifications that must be met in order to operate as a publicly traded REIT include:

- Investment of at least 75% of total assets in real estate
- Derivation of at least 75% of gross income from rents from real estate property, interest on mortgages financing real estate property, or from sales of real estate
- Operate as an entity that is taxable as a corporation
- Management by a board of directors or trustees
- Have a minimum of 100 shareholders
- Have no more than 50% of its shares held by five or fewer individuals

Aside from the obligatory dividends, REITs are an attractive investment for several other reasons. Since their inception, REITs have yielded annualized returns of up to 10-12%. As of 2015, equity REITs (the best performing sector) have generated compounded annual returns of 12%. This is 1.7% greater than the annualized return of the S&P500 and 3.1% greater than the NASDAQ (Krewson-Kelly & Thomas, 2016). Beyond returns, REITS also offer a unique way to diversify an investment portfolio. Stocks, bonds, and real estate are the three fundamental investment asset classes. By building a portfolio with diversified exposure to different asset classes, investors can minimize their exposure to market volatility while retaining anticipated returns.

Figure 1

Table 2.1 Comparative Compounded Total Annual Returns

Timeframe*	All REITs	All Equity REITs	S&P 500	NASDAQ [†]	DJIA
2015	2.3%	2.8%	1.4%	7.0%	0.2%
3-Year	10.3%	10.6%	15.1%	19.8%	12.7%
5-Year	11.6%	11.9%	12.6%	14.9%	11.3%
10-Year	6.9%	7.4%	7.3%	9.7%	7.8%
15-Year	10.8%	11.1%	5.0%	4.8%	5.8%
20-Year	10.3%	10.9%	8.2%	8.1%	8.8%
30-Year	9.4%	10.7%	10.4%	9.6%	8.4%
40-Year	12.0%	13.7%	11.4%	11.0%	7.8%
1972 – 2015	9.8%	12.0%	10.3%	8.9%	7.0%

Sustainable Investment Overview

Sustainable investing, also called Environmental, Social, Corporate Governance (ESG) Investing, has become increasingly popular in the last two decades. As issues surrounding sustainability have become more widely recognized, so too has the recognition that sustainable initiatives and financial investments go hand in hand. In 1995, \$639 billion of all professionally managed assets were under ESG investment strategies. By 2018, that number had risen to \$11.6 trillion (Conmaker & Madsbjerg, 2019). Historically, many investors thought of sustainable investment as a tradeoff; investing in companies that “do good” would inherently lead to reduced returns. The growing prominence of investments that do good *and* return well has dramatically altered that notion.

This paradigm shift has made ESG investing an increasingly popular idea, with all trends pointing to a further broadening of this investment behavior. In a newsletter released to clients, UBS's Investment Research Unity posited that "much of this growth has been organic, driven by demand and new business opportunity" (UBS, 2020). Numerous factors are driving this growth. Green technologies like wind and solar are becoming more economically viable, hedge funds are considering ESG factors as a measure of long-term performance, and investors are increasingly recognizing the interdependence of financial returns with ESG wellbeing. As the space continues to expand, so too will the drivers of its expansion.

An important aspect of ESG investing is that it does not simply include companies or industries that are green by nature, such as renewable energy or social welfare companies. Instead, any company or industry that can present a viable case for its own sustainable operations can qualify as a socially responsible investment.

This idea of integrating sustainability practices into financial investments originated in Europe, where assimilation of the investment world with social and environmentally driven initiatives continues. In 2018, the European Commission developed an action plan to finance and foster this sustainable growth (Sucher, 2012). Some notable points from this plan include:

- Establishment of a classification system for sustainable activities
- Creating an EU Green Bond Standard and labels for green financial products
- Incorporating sustainability in financial advice
- Strengthening sustainability and disclosure accounting rule making (Commission Action Plan, 2018).

Given the rate of growth of ESG investing in the United States, it is likely that the United States will adopt similar measures. Blackrock and Vanguard, both of which are prominent fund management firms in the North American market, have issued favorable statements regarding their stance in ESG investing and offer numerous products that meet this demand from their clients (Fink, 2020). Currently, there is confusion that the concepts of sustainability by which many investments are graded are unstandardized and loosely defined (Ho, 2020). The concept of ESG investing is still relatively new and holds a lot of room for development. Just as the United States has followed the lead of Europe into the sphere of sustainable investment, it is reasonable to believe that the nation will follow the European model to continue to enhance the sustainable investing industry's development and uphold its credibility.

Consumer demand for increased corporate transparency has been on the rise for many years (Global Impact Investing, 2018). So too has consumer demand for corporations to employ more sustainable practices, as is evidenced by the increased prominence of sustainability-oriented marketing campaigns in recent years. Corporations that satisfy this demand have the potential to not only increase demand for their products, but also their stock. Firms with higher sustainability ratings have increased exposure to the growing market of ESG investing.

For its part, Hannon Armstrong occupies a unique space in the field of sustainable investments. Many companies value the ESG ideas and consider them essential to their business model. Many others employ some ESG principles to gain recognition and help their overall ESG rating or give them some positive marketing material. However, in most circumstances, having an ESG focus does not change the overall structure or mission of these companies. Hannon Armstrong is unique in that it is a firm whose principal focus is on curbing global climate change. The entire premise of the business revolves around the ambitious goal of significantly reducing global greenhouse gas emissions and increasing resiliency to climate change through the funding of infrastructure, renewable energy projects, and more. This unwavering focus on sustainable investments as the core principle of corporate existence differentiates them, even within the ESG space. Investors in Hannon Armstrong can capture healthy returns on their investments while simultaneously working to improve livelihoods and communities in the long term.

Five Forces Analysis

Michael Porter's Five Forces Analysis is a long-standing model of business analysis that tries to explain how levels of profitability within a given industry can be sustained. An understanding of the five forces uncovers the roots of profitability within an industry and reveals how profits can be sustained over time (Porter, 2018). The following is Five Forces analysis of the REIT industry and considers Hannon Armstrong's place therein.

Competition

The REIT industry in the United States is highly competitive, with no single entity controlling a dominant place in the overall market share. Much like the agricultural industry, the sheer size and diversity of the real estate market is inhibitive of any single entity retaining a dominant stake. Concentration within the industry is expected to rise. The number of NYSE-listed REITs has declined since reaching a high in 2006. This is largely a result of the subprime mortgage crisis of 2008, which shook many overleveraged players out of the industry. Many of these REITs ended up merging with larger, more stable firms or be acquired by major financial institutions (Masters, 2020). The financial crisis and reshaping of the retail industry will also undoubtedly drive further concentration of the industry, though it is too early to determine to what degree.

Hannon Armstrong's niche specialization within the REIT industry currently leaves it without direct REIT competitors. However, the company is threatened by several different competitors, which are discussed in the Strategic Analysis section of this report.

Potential of New Entrants in Industry

Due to the high level of capital needed to start up, the potential for new entrants into the industry is low. Furthermore, the growing prominence of online commerce and increased levels of people

working from home will disincentivize nascent players from entering the industry. We believe that these factors will drive entrepreneurs and their capital away from the retail and office sectors, two of the most prominent REIT markets.

Contrary to the industrial trend, we believe Hannon Armstrong is highly threatened by new entrants into their industry. The renewable energy sector is expanding rapidly, as is the public call for projects that remediate environmental damage. Hannon's Armstrong's consistent year-over-year growth and profits will certainly attract new players to their specialized category in which Hannon Armstrong currently operates.

Power of Suppliers

Standard REITs are supplied by a high number of different factions, including construction companies, maintenance service providers, investors, and many more. Given the high level of competition that exists amongst all these supply factions, the threat that any single supplier threatens the profitability of a REIT is low. If one supplier were to pose a threat to the industry, it would be investors. Given the liquid nature of REITs, an economic fallout or widespread need for liquidity could cause investors to rapidly divest and leave a REIT without capital. This very phenomenon took place in the early weeks of the COVID-19 pandemic.

Hannon Armstrong is supplied by producers of renewable energy equipment, construction companies, and investors. The power of suppliers to Hannon Armstrong parallels that of the wider REIT industry outlined above as there is a high degree of competition amongst suppliers.

Power of Customers

The customers of REITs are as diverse as the sectors in which REITs operate. The customers of most retail and industrial REITs are the businesses that rent space within their facilities. The customers of residential REITs are the developers of the real estate, or home buyers themselves depending upon how the REIT is structured. Broadly speaking, customers of residential REITs hold little influence the industry as the need for housing is constant and invariably necessary. In the retail and office sectors, however, customers hold more power over the industry as the capacity to conduct business online has given these customers an alternative to the brick-and-mortar businesses they have traditionally operated.

We believe that the customers of Hannon Armstrong hold a moderate degree of power over the company. Customers are drawn to Hannon Armstrong because of their knowledge and experience within their industry, but they could turn to traditional banks to finance their projects. Some installers of renewable energy infrastructure also offer financing programs. If Hannon Armstrong can continue to compete with these alternatives, there is no reason that a customer would cease to the firm's expertise.

Threat of Substitute Products

The most immediate threat to the REIT industry is a widespread transition to online commerce and business, which has been detailed above. Less immediate is the threat that alternative means of financing pose to the industry. Though the deployment of a feasible substitute is years away (in both a technological and regulatory context), blockchain technology has the potential to

dramatically alter the way REIT-like initiatives are funded and managed. A full discussion of blockchain's potential to decentralize industries is beyond the scope of this report. In very simple terms, blockchain technology could enable investors to cut out the middleman between them and the real estate projects that generate their returns. REITs are that middleman, and the industry would do well to reconsider how they add value to the services they provide.

Strategic Analysis

Mission, Purpose, and Values

Hannon Armstrong's mission and purpose lie in their belief that every investment should improve our collective climate future, and thus the company requires that all their prospective investments be neutral or negative to carbon emissions or be otherwise beneficial to the environment. In short, their aim is to "make climate-positive investments with superior risk-adjusted returns" and for "to improve our climate future" (Investing in Climate, 2020).

Business Units

Hannon Armstrong invests capital into a wide variety of projects that generally fall into three distinct markets: behind the meter energy projects, grid-connected projects, and sustainable infrastructure projects. Behind the meter energy projects focus on reducing energy usage and/or cost through solar technologies or overall energy efficiency improvements. This category accounts for about 59% of Hannon Armstrong's portfolio, and investment examples include public-private partnerships supporting universities' low carbon campus initiatives, community and residential solar, and microgrid installations in places such as a Marine Corps depot.

Complimenting Hannon Armstrong's BTM investments are their grid-connected projects which make up 40% of their portfolio and are primarily focused on the wholesale power markets. They invest heavily in onshore wind, and land solar projects including the equity ownership of over \$500 million in renewable generators in partnership with companies such as Amazon, Walmart, and Xcel Energy (BlackRock in U.S., 2020).

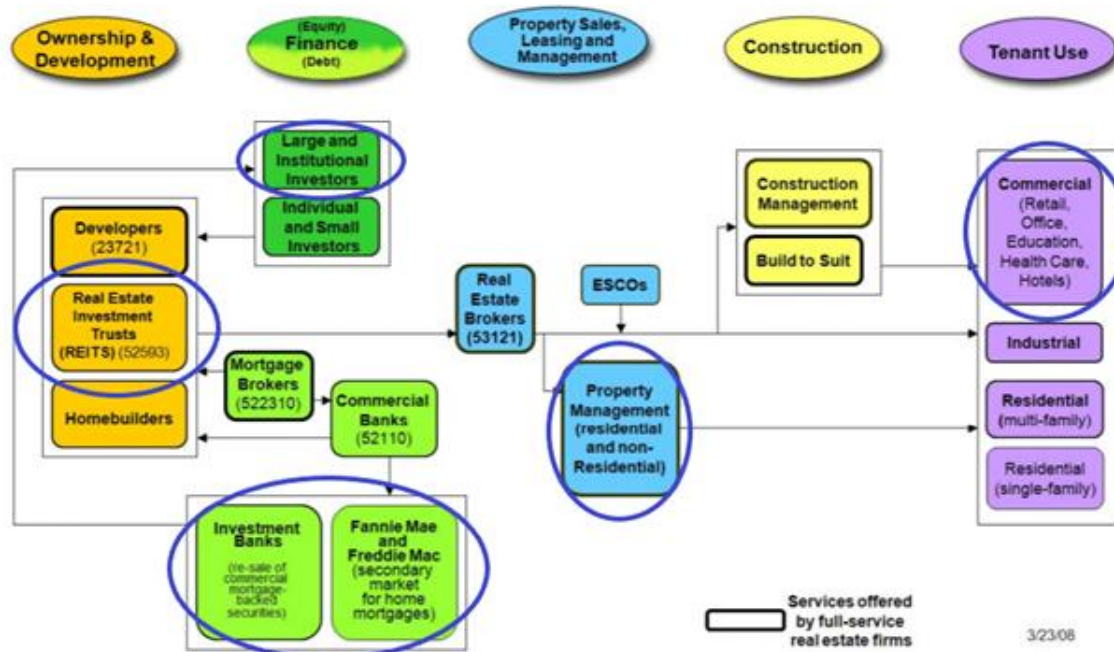
Sustainable infrastructure projects make up the last 1% of Hannon Armstrong's portfolio, and aim to positively affect the environment through upgrades in infrastructure and efficient use of natural resources. Investment examples in this market include stormwater remediation and ecological restoration, as well as the privatization of a utility, which allowed for equipment and distribution system upgrades. As showcased by their diversified portfolio of projects, Hannon Armstrong is considered a specialized REIT which invests in projects that propel its mission.

Alongside their BTM and grid-connected investments lies the company's sustainable infrastructure endeavors which consist of projects aimed at improving efficiencies and mitigating environmental impacts. Examples of their infrastructure improvement projects include ecological restoration, stormwater remediation, and a \$200 million utility privatization project for the purpose of increasing electricity distribution efficiency and <\$20 million in ecological restoration projects (Parnassus Investments, 2020). Along with investing directly into environmentally focused projects, Hannon Armstrong has issued \$4.6 billion of green debt since 2013. As part of the Nasdaq Sustainable Bond Network, Hannon Armstrong's bonds are dedicated to green projects whose purpose is to reduce GHGs or improve the environment by other means (Macerich, 2020).

Place in Value Chain

Placing REITs into a specific box within the value chain is not an easy task due to the nature of the real estate industry and all the constituents involved. The figure below from Marcy Lowe and Gary Gereffi's Analysis of the U.S. Real Estate Value Chain, helps to illustrate the complexity of this matter (Lowe, 2008). The areas circled in blue indicate primary categories along the value chain in which REITs are involved. Of course, this figure only reflects the real estate-related pieces of the value chain. Arguably, Hannon Armstrong is also heavily involved in the primary activities of operations and services, and the secondary activities of procurement and technological development according to Michael Porter's value chain philosophy (Tarver, 2020).

Figure 2



Primary Target Market(s)

The primary target market for Hannon Armstrong is investors and stockholders who are looking to invest in energy efficiency projects or sustainable infrastructure, as well as cities and organizations who can benefit from those investments.

Value Proposition to the Customer

Hannon Armstrong is a specialized REIT that not only prioritizes returns for investors, but also adds values to consumers and projects by investing in climate negative or neutral projects that help society move toward a sustainable and resilient future. "Thanks to our investors, customers, partners, and employees, Hannon Armstrong is proud to stand at the forefront of the transition to a low-carbon economy" (Hannon Armstrong Corporate, 2020). This speaks strongly to the types of customers whom Hannon Armstrong works and seeks to attract. A large part of the value that is embedded in their services lies in the differentiation of their investment strategy, which works toward sustainable solutions and transitioning the economy away from fossil fuel dependence.

Detailed Analysis of Major Competitors

Hannon Armstrong occupies a unique position within the market, and as such, potential competitors include not only other REITS, but also utilities, yieldcos, and even conventional private equity firms, banks, or life insurance companies. For the sake of this analysis we will examine competitors that operate in a variety of these spaces, and that tout sustainability as one of their major tenets.

BlackRock: BlackRock is a large investment firm that deals in everything from stocks, bonds, real estate funds, retirement funds, and more. In recent years, BlackRock has made sustainability a much larger part of their operations and how they attract investors. Just this year, BlackRock committed to ensuring that any company they invest in has thorough sustainability disclosures. Not only that, but BlackRock's CEO, Larry Fink, stated that the company will vote against management and board directors of companies that are not making sufficient progress on sustainability related goals. While BlackRock is a well-known company to many, and undoubtedly attracts many "conventional" investors, increasingly it seems that they are a one stop shop for people who are looking to grow their wealth and support ESG initiatives the world over. Interestingly, in February 2020, BlackRock disclosed a 9.1% ownership stake in Hannon Armstrong which will likely affect the operations of both organizations (Simply Wall St., 2020). BlackRock's net income in 2019 was \$4.5 billion (BlackRock in U.S., 2020).

Parnassus Investments: Parnassus is an investment firm that has committed to investing in high-quality companies, which they define in part as businesses that have sustainable competitive advantages and positive ESG profiles. Parnassus uses a comprehensive screening process to consider the environmental record and social impact of every company they work with. The company has explicitly stated that they work with "responsible investors" and as of September of 2019 they adopted a firm-wide policy of avoiding investments in fossil fuel companies altogether. Parnassus actively divests from companies that have associations with activities that detract from positive climate action and sound social standards. They have five mutual funds that they offer to investors, each with a unique mix of investments, depending on the investor's goals. Parnassus Investments' 2019 net income information is unavailable (Parnassus Investments, 2020).

Macerich: Real estate investment trusts are increasingly turning to sustainability as a means of attracting investors, and Macerich is no exception. The company has declared itself "the premier, pure-play, high-end mall REIT" while simultaneously positioning itself as a sustainability leader in the REIT space (Macerich About, 2020). Macerich was voted as one of America's most sustainable companies by Barron's, they rank within the top 30 of the EPA Green-Power Partnership, and they achieved the #1 Global Real Estate Sustainability Benchmark (GRESB) ranking in the North American Retail Sector for five straight years (Macerich About Sustainability, 2020). Additionally, four of their malls are LEED certified, and one of their buildings boasts the largest fuel cell energy system in U.S. mall history. Every shopping center that Macerich purchases is subject to an environmental site assessment. Overall, Macerich is performance-focused, and while they clearly have made efforts to make sustainability a central part of what they do, it seems their target market is still traditional investors who are looking for solid returns. It is

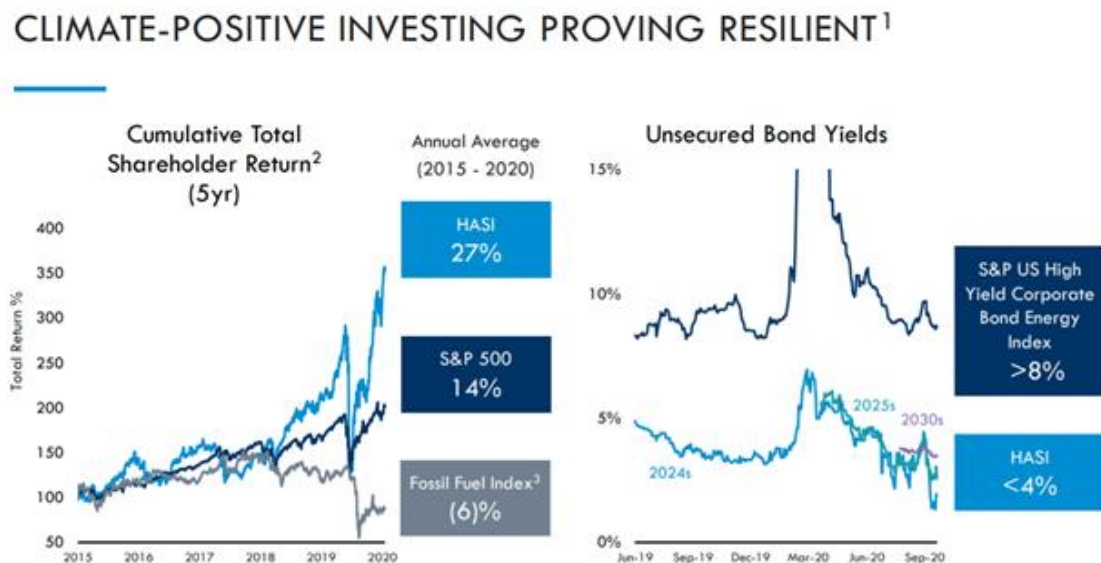
important to note that Macerich reported a net loss of over \$22 million in the third quarter of 2020, an unsurprising fact given the rising popularity of online shopping, and more prominently, the effects of COVID-19 on retail businesses this year. Macerich's net income in 2019 was \$102 million. (Macerich Leasing, 2020).

Hercules Capital: Hercules is another avenue of investment for sustainability-minded investors. Hercules Capital is a business development company that focuses on venture development for innovative entrepreneurs. They primarily work within the life sciences, technology, and sustainable and renewable energy sectors. Within the Sustainable + Renewable Technology vertical, Hercules targets disruptive companies in the categories of vehicle technology, battery technology, agriculture, and more. Hercules states that they “take a comprehensive approach to integrating environmental, social and governance (ESG) criteria into our investment process.” (Governance Documents: Hercules, 2020). Although sustainability is part of what Hercules invests in, their business model is not entirely built around it, and their investors may or may not choose to invest with Hercules for this particular factor. Hercules Capital's net income in 2019 was \$173 million.

Differentiation

Hannon Armstrong differentiates itself primarily by positioning itself as the first and only public company in the U.S. that is totally dedicated to investing in climate change solutions. This separates it from other investment firms and REITs, which may have a sustainable investment sector or some focus on sustainability but are not 100% dedicated to it. Another point of differentiation that gives Hannon Armstrong an advantage over their potential competitors in the fossil fuel industry and S&P 500 is their rate of return for investors. As seen in the chart below from Hannon Armstrong's November 2020 Corporate Profile, the company's 5-year shareholder return is significantly higher than those of the S&P 500 and the Fossil Fuel Index. Additionally, the bond yields from Hannon Armstrong have been shown to be a safe and relatively stable investment.

Figure 3



SWOT Analysis

Strengths: Hannon Armstrong's major strengths include its reputation as a company that is dedicated solely to investments in climate change solutions, along with notable increases in their invested assets over the last five years and solid returns for investors. Additionally, the continual and expected growth of the sustainable infrastructure market is a huge strength for Hannon Armstrong as it indicates that interest in climate positive investments will only continue to grow.

Weaknesses: Weaknesses of the company include the possible credit risk and interest rate risks that they have taken within their portfolio to achieve their strong yields, along with the fact that there is little diversity among the top leadership of the company. Though Hannon Armstrong is focused on sustainable investments, in an era where many consumers and investors demand higher standards for the triple bottom line, the social aspect of diversity within company leadership should not be ignored.

Opportunities: Opportunities for Hannon Armstrong include the ever-increasing market demand for renewable energy infrastructure, as well as the ongoing shift of capital from traditional fossil fuel investments to ones that are sustainable and forward thinking. As a financier of projects pertaining exclusively to the curtailing of global climate change, Hannon Armstrong is uniquely positioned to consult with companies seeking to improve their ESG rating, finance their initiatives, and facilitate the implementation of their projects. Moving forward, the company may want to look into broadening their services and target market to meet this demand. Currently, they only provide services to leading companies in energy efficiency. However, the rapidly expanding pool of corporations seeking to offset their carbon emissions presents a new target market with high growth potential. Modifying their services to fit the needs of this market would not be too dramatic of a transition from the services that Hannon Armstrong already offers.

Threats: Threats to Hannon Armstrong's continued success and growth include the increasing number of investment firms and REITs that emphasize sustainability. Over time, this will likely make it more difficult for the company to stand out amongst many socially- and environmentally conscious investment options. Additionally, Hannon Armstrong has grown considerably in the last several years, it is generally more difficult to generate the same high returns when working with larger amounts of capital.

Blue Ocean Strategy Analysis

Arguably, Hannon Armstrong is a blue ocean company in a space that has the potential to become a red ocean before long. Kim and Mauborgne explain that red oceans depend on competition for players to survive, whereas blue oceans present opportunities for vast growth because there is not yet any competition within the space (Kim, 2005). As mentioned above, the overall shift of the economy and values to more sustainable and climate positive investments does present a threat to the unique market position that Hannon Armstrong currently occupies, which means that they will need to continue to differentiate themselves and the impact they make for potential investors. However, it's important to note here, that unlike commodity-based industries or markets, we believe that a more crowded sustainability space will have overall positive implications for the ultimate stakeholder of this sector: the environment itself. From a company perspective however, Hannon Armstrong will need to use the four actions framework

to consider which factors to raise, reduce, eliminate, and create in order to continue to provide outstanding value to their shareholders.

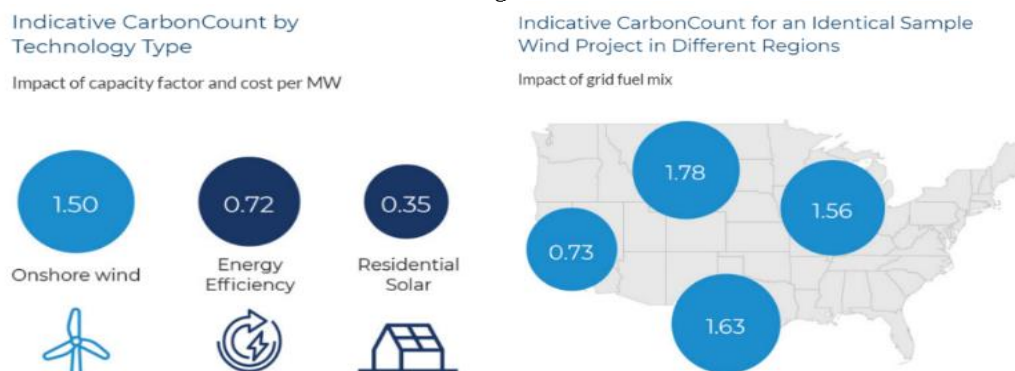
Impact Analysis

With annualized returns of 25% over the past 5 years, Hannon Armstrong's commitment to climate-positive investments has provided an example of a company doing well by doing good, and it is well positioned to continue to be at the forefront of the environmentally focused investments trend (Hannon Armstrong Form 10-K, 2020).

While Hannon Armstrong's principal focus remains on investments that mitigate, the company has also shown continued progress in the social and governance realms of responsible and sustainable business practices. The global business community appears to be moving towards a standardized method of Environmental, Social, and Governance (ESG) reporting, however, the company has proven itself a model for how to maintain transparency when communicating their goals, accomplishments, and areas that need improvement (Palmer, Gregory, & Wood, 2020). They accomplish this by utilizing internal controls, independent auditors, ratings agencies and the inclusion of extra-financial material in their financial filings.

Most notably, they have developed CarbonCount, which is a scoring tool used to evaluate investments by determining the efficiency by which each dollar of invested capital reduced annual carbon dioxide equivalent emissions (CarbonCount, 2020). This tool is particularly important for the investment sector as it is comparable across projects and it provides transparency for investors and stakeholders about the true cost of a particular investment. In addition, it promotes accountability by taking out ambiguous terminology like "sustainable," and it encourages the allocation of capital where it can make the most impact (CarbonCount, 2020). This tool positions Hannon Armstrong as a prevalent leader within their industry- a company that is driving the force in true accountability and impact.

Figure 4



Environmental Purpose

As an environmentally focused REIT, Hannon Armstrong's primary focus is on the impact that their investments and operations have on the environment, specifically with respect to the overall effect that their activities have on carbon emissions (Figure 4). Utilizing internal analysis and external auditors such as CarbonCount³, the company evaluates investment opportunities and their own business practices that most effectively reduce carbon emissions while simultaneously

bringing down costs and mitigating climate change and regulatory risk. By integrating material aspects of ESG into their financial filings, Hannon Armstrong strengthens their position in the REIT industry by proving their commitment to mitigating the risk to their assets from climate change.

As a financial firm, Hannon Armstrong's investments are far more environmentally consequential than their internal business practices. However, the company has shown a strong commitment to their values by releasing a thorough environmental metric report every year. The report showcases that their facilities are 100% powered by renewable energy and all paper used is certified post-consumer recycled. They also recycled 3 out of 4 tons of paper used in 2019. (Hannon Armstrong, 2020). Moreover, they provide metrics on Scope 1, 2, and 3 emissions for their business activities as well as indirect CO₂ emissions (607 tons in 2019) (Impact Report, 2019). In addition to their direct consumption mitigation initiatives, the company also partners with, and provides support for several environmental organizations including The United Nations Global Compact, The Task Force on Climate Related Financial Disclosures, and the Smithsonian Environmental Research Center.

Social Responsibility

While Hannon Armstrong has been a leading innovator in the environmentally focused aspects of the ESG financial segment, their internal social qualities are on par or slightly better than industry (Waters, 2020). The company's total metrics concerning diversity in their workforce are 38% women, 23% people of color, and the board includes two women compared to seven men (Impact Report, 2019). As of 2018 the company has provided annual diversity training to all their employees as well as sponsorship of the Women of Renewable Industries and Sustainable Energy (WRISE) development group. One of the most impressive social metrics for the company is that the CEO's average compensation is 23 times that of their average employee, compared to the 320 times that the average head of an S&P company makes compared to their employees (Mishel 2020). Though their diversity metrics are commendable, we maintain that opportunity remains for Hannon Armstrong to incorporate more diversity into top leadership roles at the company.

In reference to direct employee relations, Hannon Armstrong provides a wide range of benefits that go beyond performance-based assessment of compensation and their equal pay initiatives. They also offer comprehensive health insurance coverage, an employee wellness program, free legal and financial tools as well as tuition reimbursement.

From an external point of view, Hannon Armstrong is taking steps to increase their commitment to strengthen relationships with the communities in which they operate. The company has outlined a detailed Business Partner Code of Conduct that they use to screen and periodically assess their investment partners. This code of conduct includes a broad set of criteria by which partners must abide, and includes ethical, legal, and environmental requirements (Corporate Governance, 2020).

Figure 5

Governance: Accountability and Transparency



At the core of the value proposition that Hannon Armstrong provides to its stakeholders is a corporate structure dependent upon transparency and accountability. Recognizing the rapidly changing regulatory environment and the increasing interest from conscientious investors, Hannon Armstrong has established a committee that is charged with assessing and evaluating ESG initiatives and objectives within the company. The Nominating, Governance, and Corporate Responsibility Committee, which reports to the board, assesses the environmental impact of the company's investments, the metrics used to assess these investments, and the progress made in all ESG goals for the purpose of making recommendations to management.¹¹ Along with designing a corporate structure that fully integrates ESG principles in all levels of decision making, Hannon Armstrong also ties all employee compensation, including at the executive level, to ESG performance metrics. As the global regulatory environment continues to shift towards inclusion of extra financial analysis in accounting disclosures, Hannon Armstrong has positioned itself well to adapt to these changes.

Overall Impact

Hannon Armstrong's emergence as a leader in the ESG-focused financial industry, specifically concerning their innovative use of the REIT, has been recognized by a number of highly credible institutions (Esposito, 2020). Managing Director Susan Nickey received Woman of the Year for 2020 for her outstanding work in the cleantech industry, the company was declared the Best ESG Sustainable Investment by CFI, and they were named one of America's Fastest Growing Companies in 2020 just to name three of the 20+ awards since 2015 (Awards and Recognition, 2020). By instilling a set of values throughout the enterprise that are in line with their sustainability goals, Hannon Armstrong has been able to effectively utilize their capital for the purpose of reducing carbon emissions across various sectors to the tune of 3.2 metric tons of CO₂ in their 2019 disclosure (Figures 6&7).

Figure 6

Carbon Reduction

Cumulative metric tons of CO₂ avoided annually

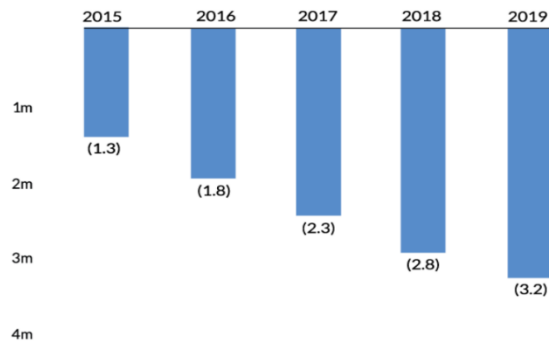
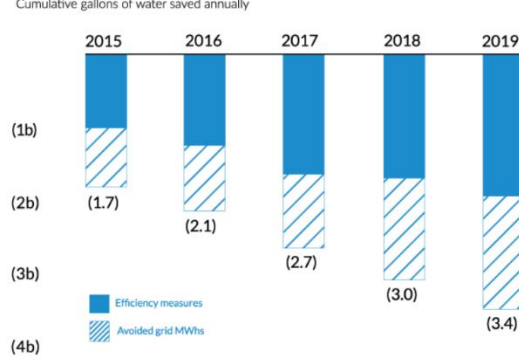


Figure 7

Water Savings

Cumulative gallons of water saved annually



Financial Analysis

While Hannon Armstrong currently has about \$2.1 billion of invested assets held on its balance sheet, they also manage capital for other parties by underwriting loans, managing debt portfolios, and brokering transactions for a fee (Hannon Armstrong, 2020 10K). These are typically held in securitization trusts outside of Hannon Armstrong's ownership and therefore, not reported on the balance sheet. In 2019 these assets represented about \$4.1 billion in managed capital for Hannon Armstrong fee collection (Hannon Armstrong, 2020 10K). The company reports its invested assets in the following categories:

- Equity investments (in either preferred or common structures in unconsolidated entities) (Hannon Armstrong, 2020 10K)
- Government and commercial receivables or securities (such as loans for renewable energy and energy efficiency projects) (Hannon Armstrong, 2020 10K)
- Real estate (such as land or other assets leased for use by sustainable infrastructure projects typically under long term leases) (Hannon Armstrong, 2020 10K)

As shown in Exhibits 2 & 3, the asset mix has stayed largely consistent over the past five years with around two thirds of assets being held in securitization trusts (likely their least profitable pool with management fees in the 1-2% of AUM range), around 20% of assets in loan receivables, and the rest split between equity and real estate investments (see Exhibits 1, 2, & 3). The most noticeable change in this mix comes from the growth in commercial receivables at the expense of the government receivables investment pool. This is likely due to the declining interest rates forcing the company to reach further into the commercial market for more yield (average interest investments receivables have grown about 2% to a little over 7.3% of receivable assets, see Exhibit 4).

Exhibit 1: Investment Mix table (all dollar amounts in millions)

(in millions)	2019	2018	2017	2016	2015	2014
Equity Method Investments	\$ 499	\$ 471	\$ 523	\$ 363	\$ 319	\$ 144
Gov. Receivables	\$ 263	\$ 497	\$ 519	\$ 526	\$ 401	\$ 284
Commercial Receivables	\$ 896	\$ 447	\$ 473	\$ 516	\$ 383	\$ 267

Real Estate	\$ 362	\$ 365	\$ 341	\$ 172	\$ 156	\$ 114
Securitization assets	\$ 4,176	\$ 3,504	\$ 2,880	\$ 2,356	\$ 1,929	\$ 1,800
Total Managed assets	\$ 6,196	\$ 5,284	\$ 4,736	\$ 3,933	\$ 3,188	\$ 2,609

Over this 5-year period, Hannon Armstrong has more than doubled its invested assets, showing an average growth rate of 18.89% annualized over the period (Exhibit 1). In 2014, the company held or controlled around \$2.6 billion in assets compared to its current asset pool of \$6.1 billion (Exhibit 1). This growth was done while keeping debt relatively low. Debt to assets moved from 1.37 in 2014 to 1.65 in 2019 (Exhibit 6). This resulted in an increase in debt to assets of around 3.75% annualized- which was far outpaced by growth in assets, and profitability during this time (Exhibit 6).

Exhibit 2: Asset Mix pie chart, 2019
(dollar amounts in millions)

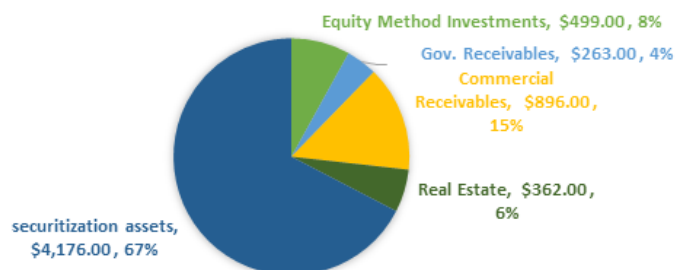
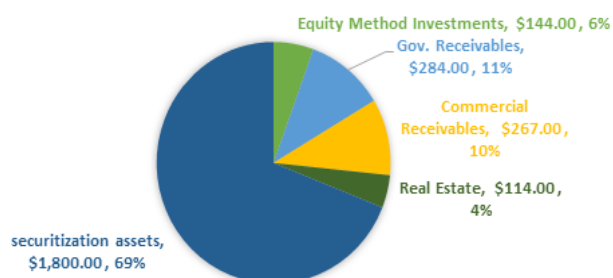


Exhibit 3: Asset Mix pie chart, 2014
(dollar amounts in millions)



This growth has shown itself to be extremely profitable in the long term as well. Breaking down Exhibit 4, Hannon Armstrong has managed to grow its average yield from 5.5% in 2014 to 6.9% in 2019. This growth was driven by an increase in interest receivables (much of which is likely from securitized assets) as well as yield growth from the company's direct investments in real estate. The company grew sales at an impressive rate of 25.61% annualized, only to be outpaced by a staggering 52.87% annualized growth in net income and 43.59% annualized growth in free cash flow (Exhibits 5, 6).

Exhibit 4: Investment yield table

Year	2019	2018	2017	2016	2015	2014
Avg. Interest receivables	7.3%	6.8%	5.3%	5.6%	5.3%	5.4%
Avg. Interest investments	4.3%	4.1%	4.2%	4.1%	5.3%	5.7%
Avg. Real estate yield	7.1%	7.0%	7.0%	7.3%	6.5%	6.3%
Avg. Yield	6.9%	6.5%	5.6%	5.8%	5.5%	5.5%

Aside from an increase in debt levels (liabilities tripled over the 5-year period), Hannon Armstrong shareholders also suffered significant dilution during this time (Exhibit 5). The company's share count more than quadrupled over the past five years, showing that the company has leaned on shareholders as well as borrowings to fund this growth both in revenues and profitability (Exhibit 5).

Exhibit 5: Income/Profitability table (dollar amounts in thousands)

Year	2019	2018	2017	2016	2015	2014
Net revenue	\$ 141,581	\$ 139,396	\$ 105,987	\$ 81,198	\$ 58,679	\$ 45,275
Interest expense	\$ 64,241	\$ 76,874	\$ 65,472	\$ 45,241	\$ 26,385	\$ 16,655
Income tax expense	\$ (8,097)	\$ (2,144)	\$ (885)	\$ (141)	\$ (118)	\$ (26)
Net income	\$ 81,564	\$ 41,577	\$ 30,856	\$ 14,756	\$ 8,034	\$ 9,770
Cost of goods sold	\$ 43,470	\$ 40,742	\$ 30,884	\$ 27,170	\$ 24,044	\$ 16,368
Gross profit	\$ 98,111	\$ 98,654	\$ 75,103	\$ 54,028	\$ 34,635	\$ 28,907

Overall, this strategy seems to have paid off for Hannon Armstrong shareholders, with their results reflected in growth in earnings, dividends, and ultimately stock price. In 2014, the company was losing \$0.68 per share in earnings (Hannon Armstrong, 2015 10K). Since then, the company has grown earnings in four of the past five years, producing earnings of \$1.24 per share in 2019 (Exhibit 6). Hannon Armstrong managed to grow its dividend in all these years, boasting a dividend growth rate of 7.81% annualized since 2014 (Exhibit 6). Return on equity averaged 5% over the period with a record return of 9.4% in 2019 (Exhibit 6). Ultimately, investor satisfaction has resulted in shares growing at a rate of 25.13% annualized from \$10.11 per share in 2014 to \$31.01 per share in 2019 (Exhibit 6). As of 2019 its shares were valued at a reasonable 25 times earnings, below the stock's average price to earnings of 37 over the period. From a price to earnings perspective, Hannon Armstrong's stock would be considered statistically cheap at the end of 2019 compared to its 5-year average (Exhibit 6).

Exhibit 6: Shareholder Information Table

Year	2019	2018	2017	2016	2015	2014
Market price	\$31.01	\$17.52	\$20.70	\$15.18	\$14.41	\$10.11
Earnings per share	\$1.24	\$0.75	\$0.57	\$0.32	\$0.21	\$0.43
Outstanding shares	64,771,491	52,780,449	50,361,672	37,010,603	30,761,151	15,892,927
Dividend per share	\$1.34	\$1.32	\$1.32	\$1.23	\$1.08	\$0.92
Free cash flow (thousands)	\$115,895	\$129,782	\$79,951	\$106,336	\$50,110	\$18,987
Debt to assets	1.65	1.60	1.40	1.49	1.42	1.37
Price to earnings ratio	25.01	23.36	36.32	47.44	68.62	23.51

Hannon Armstrong's 25.13% annualized return handily beats out both the overall 5-year return of the REIT sector (5.2%) and the S&P 500 (9.9%) (Matt Frankel C, 2020). To put that return into perspective, \$10,000 invested in Hannon Armstrong in 2014 would be worth a little over \$30,000 today (Exhibit 8). That same amount put into the broader REIT sector would be worth almost \$13,000 and the same amount put into the S&P 500 would be worth about \$15,500 (Exhibit 8). This would seem to indicate that management at Hannon Armstrong has been delivering on their objective of growing profitable returns with predictable cash flows.

Exhibit 8: Growth of \$10,000 table

Year	2019	2018	2017	2016	2015	2014
Hannon Armstrong	\$30,672.60	\$17,329.38	\$20,474.78	\$15,014.24	\$14,253.21	\$10,000
REIT Sector	\$12,884.83	\$12,247.94	\$11,642.53	\$11,067.04	\$10,520.00	\$10,000
S&P 500	\$15,691.73	\$12,175.46	\$12,985.77	\$10,874.04	\$9,927.00	\$10,000

A good portion of this growth is likely due to the growth of the securitized trust assets that currently make up around two thirds of the portfolio. Since these are mostly fees taken from managing portfolios of debt instruments, it is likely that much of this performance is based around a similar growth trend in bond portfolios due to sustained low interest rates. This is also evidenced by the dramatic shift in receivables coming from commercial sources as opposed to government receivables. While it is quite likely there has been less opportunity to invest in government projects with climate impacts given the current administration, this shift to commercial receivables is likely also related to a reach for higher yield projects as well.

Another factor in Hannon Armstrong's outperformance of the market could be the effect of the recent stock market decline related to the global pandemic. Effects were widely felt across industries and although the market has recovered some of those losses later in 2020, there is still some room to make up the difference in previous year gains. The S&P 500 performance has probably eaten into this difference since then, but it is likely not significantly. This return has also outpaced inflation on U.S. home prices significantly- which has averaged between 2% and 3% over the last five years. (Inflation Rate for Housing between 1967-2020). All this points to the ability of Hannon Armstrong management to invest in growing areas of the market that seem to be able to command high yields. The type of growth Hannon Armstrong has seen more closely resembles that of the renewable energy sector (which has grown 100% from 2000 to 2018) (Renewable Energy, 2020). An increase in demand for renewables combined with more and more apparent savings from energy efficient building upgrades is clearly driving an increase in demand for the type of projects that Hannon Armstrong provides funding to.

Hannon Armstrong seems to be consistently delivering value to its stakeholders by investing in low carbon projects while providing predictable cash flows. The only area of concern is properly evaluating what sort of risks are being taken to achieve these yields. Hannon Armstrong has steadily increased its yield from receivables during a time when interest rates have generally declined or stayed flat. The portfolio could be exposed either to credit risk, interest rate risk, or both to varying degrees. This could be due to their debt investments increasing in value because of these rate changes. Average interest from investments still sits at 4.3% which is more than double the rate of government bonds. This discrepancy may point to higher risks being taken by the company to create these returns (Exhibit 4). The interest receivables number may also showcase some sensitivity to rising interest rates in the future if it is based, in part, on fees collected as a percentage of assets under management or investment performance. A rising interest rate environment in the medium to long term seems likely and investors will not truly know the exposure Hannon Armstrong has in this area until that occurs. The other challenge management faces are in scale as high returns become more difficult with larger pools of capital.

Conclusion and Recommendations

Overall, Hannon Armstrong is a leader in the impact investing space. As a specialized REIT, they have a unique competitive advantage through their utilization of green energy projects which position the company well in the growing sector of renewables in addition to the more traditional REIT space. This core operations strategy may indicate that the firm is operating within a Blue Ocean, with little direct competition to hinder their performance.

In addition, the company's use of yearly impact reports, third party audits, greenhouse gas tracking for Scope 1, 2 and 3, as well as their governance structure surrounding ESG goals showcase their transparency. Hannon Armstrong is not in the business of greenwashing and is held accountable through third party audits in their environmental claims as well as to their investors and the general public. Direct connections with business operations and aspirations to the UN's Sustainable Development Goals has become a standard practice for socially and environmentally conscious firms and Hannon Armstrong is no different. In its 2019 Impact Report, the company directly linked its operations to five SDG goals mostly surrounding access to clean energy and climate action. Hannon Armstrong's environmental practices, reporting, and transparency are certainly above par for any company and no recommendations can be given on their ability to continue making an impact in this space.

Some critics have pointed out that wind and solar projects do not appreciate like traditional real estate properties, and therefore, the long-term return viability of these projects could become more limited, leading to losses in the future (Trefis, 2020). To help reduce this risk, it may be prudent for Hannon Armstrong to diversify its portfolio to include more buildings or real estate. While keeping in line with its own mission and values, however, investments could be made in only the most environmentally friendly projects including those certified with LEED, Energy Star, or the National Green Building Standard. Investments like these may help lower risk, increase returns, and diversify their overall impact. It is worth noting, however, is that Hannon Armstrong's lack of traditional real-estate investment may result in an increased ability to sidestep the devastating effects of COVID-19, which are suspected to decrease demand for commercial real-estate and lead to vacancy rate increases of 2% annually (Masters, 2020). Obviously, there are no clear-cut solutions in investment strategies and Hannon Armstrong's professional finance team is expected to continue to rise to the occasion in managing this risk.

As described in the Finance Analysis section, Hannon Armstrong has a strong position, and their financials point to continued growth and success in the coming years. From an industry point of view, the election of Joe Biden in 2020 suggests the potential for even stronger growth. The president-elect aims to eliminate carbon emissions from the power sector by 2035- a goal which would lead to unprecedented creation of renewables (Eckhouse, 2020). This hopeful industry outlook, in addition to an unstable commercial real estate market due to COVID-19 means Hannon Armstrong's strategy in investing heavily in renewable infrastructure will likely continue in the coming years.

Unlike their strong competitor (and part owner), BlackRock, Hannon Armstrong does not have a transparent or robust corporate political responsibility strategy. Most notably, BlackRock publishes *ViewPoints*, a series of papers released that illustrate their position on topics ranging from Black Lives Matter to ESG reporting standards for the industry (Viewpoints, 2020).

Hannon Armstrong should take a page from their book and better improve their own transparency and advocacy for policy that aligns with their own mission and values. Despite a strong reputation within the REIT industry, Hannon Armstrong has very little involvement pushing policy and instead, focuses a lot of their energy on internal accomplishments. CEO Jeff Eckel stated “this country has a carbon epidemic. It is going to be a devastating long-term impact and we have to get at the fix now” (Investing in Climate, 2020). If this is a core belief held by the company, more thought should be given to expanding their own CPR actions in order to push industry standards and transparency into a new frontier.

Lack of industry standards in Environmental Social and Governance investment measurements are a major setback. Competitors to Hannon Armstrong have published extensive arguments in weighing the pros and cons of a variety of standards as well as personal suggestions for investment companies to adopt on a large scale. Although arguably a leader in measuring the CO2 impact of investments thanks to their innovative CarbonCount tool, Hannon Armstrong has yet to publicly push for ESG reporting standards. We suggest that Hannon Armstrong leverage their impressive innovations to construct an industry standard in order to strengthen their own reputation and the accountability of other firms.

From a social aspect, Hannon Armstrong may be falling below its competition when we look at issues such as Black Lives Matter. Following police brutality this summer and the subsequent protests, Hannon Armstrong expressed “solidarity with our black employees and the black community” (Armstrong, 2020). Although this statement and the charitable donations given by the organization are a step in the right direction, they fall short of decisive internal action as well as external political expression. In response to this same crisis, BlackRock announced that it would increase the number of Black employees in their company 30% over the next four years as well as double the number of Black senior leaders in the same time frame (Eckett, 2020). Although strong political stances may be a financial risk, Hannon Armstrong’s lukewarm statement of support for this cause indicates more that they are unwilling to take the same decisive and transparent actions that organizations like BlackRock have done. In order to strengthen the social aspect of ESG investing, Hannon Armstrong should reevaluate how they can make impact in this sector.

Lastly, despite boasting 23% of their workforce to be People of Color in 2019, we believe that Hannon Armstrong could be making more actionable efforts for internal diversity and inclusion. Zero of the board members are People of Color and it seems that most of the 23% that they are referencing are in the lower employment level. The company did reach a target 20% of women on their board in 2019, however, with seven board members, the two women fall short of more progressive targets set by California for 2021 which states that any board with six members should have three women on the board (Posner, 2020). Hannon Armstrong should commit to not only diversifying its top board and upper management with more women and more notably, with BIPOC. Creating this change will bring the organization to par with competitors and will create a greater impact for an industry that has long seen the exclusion of minority groups amongst their ranks.

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